

GOVERNMENT OF CHHATTISGARH
Public Works Department (PWD)

Office of the Engineer-in-Chief (Pramukh Abhiyanta)
Nirman Bhawan, North Block, Sector-19, Nava Raipur, Atal Nagar, District Raipur (C.G.)

**Complaint Management & Project
Tracking System (CMPTS)**

Citizen Grievance & Public Works Transparency Platform
Web Portal + Mobile Apps (Public & Internal)

Requirement Specification

Bilingual Edition — English & Hindi (हिंदी)

Field	Detail
Prepared For	Office of the Engineer-in-Chief, Public Works Department, Government of Chhattisgarh
Version / Date	1.0 (Draft for Circulation) • 19 June 2026
Channels	Web portal and mobile apps for public and internal users
Deployment	On-Premises (PWD / State Data Centre)
Languages	Hindi and English (bilingual)
Classification	Internal — For Official Circulation

This document is available in both English and Hindi (separate and combined editions).

Contents

1. Introduction & Objectives

Purpose. This document defines the requirements for the Complaint Management & Project Tracking System (“CMPTS”) for the Public Works Department (PWD), Chhattisgarh. It is the common reference for evaluation, development, testing, and acceptance, and is written to be clear to both officials and technical teams.

Background. Citizens today have little visibility into PWD projects and no single channel to raise issues with evidence; departmental teams lack a unified tool to track and resolve such issues and to keep the public informed. CMPTS addresses both through one platform — delivered as a web portal and mobile apps — with a public side for citizens and an internal side for staff, hosted entirely on PWD servers.

1.1 Objectives

1. Let citizens lodge complaints against any project with photo/video evidence, in Hindi or English.
2. Give the public transparent project information — search, division filter, and live progress (% complete with dated update messages).
3. Provide staff tools to receive, assign, track, and resolve complaints with full accountability.
4. Let administrators update project status anytime, shown instantly to the public.
5. Create projects automatically the moment a work order is released — no manual entry.
6. Support unlimited team members with role-based access.
7. Deliver web + mobile apps, bilingual, secure, deployed on PWD on-premises infrastructure.

1.2 Key Terms

Term	Meaning
Division	PWD administrative unit (usually under an Executive Engineer) used to organise and filter projects.
Project	A public work (building, road, bridge, etc.) tracked in the System.
Work Order	The formal order sanctioning a work; its release triggers automatic project creation.
Complaint	An issue raised by a citizen against a project, with optional photo/video.
Ticket ID	Unique reference issued for each complaint, used for tracking.
Milestone	A defined project stage (e.g. Foundation) contributing to % completion.
RBAC	Role-Based Access Control — permissions by assigned role.

2. Scope & Assumptions

2.1 In Scope

- Two channels: a responsive web portal and mobile apps (Android/iOS) — for both public and internal users.
- Public side (no login to browse): view all projects, search, filter by division, view live status, lodge complaints with photo/video, track one's own complaints.
- Internal side (secure login): complaint handling, project & status management, team management, administration, reporting.
- Automatic project creation when a work order is released, via integration with the work-order / works-management system.
- Complaints allowed for both in-progress and completed projects.
- Public progress view as % completion plus dated, timestamped update messages.
- Unlimited role-based team members; full bilingual (Hindi/English) interface; SMS/app notifications.
- On-premises deployment on PWD-controlled infrastructure.

2.2 Out of Scope (this phase)

- Payments, tendering, and contractor billing.
- Accounting/ERP integration beyond the work-order feed; AI-based image verification. (Possible later phases.)

2.3 Assumptions

- PWD provides the on-premises servers, network, domain, and SSL certificate.
- PWD provides an API/feed from the work-order system, plus the division list and any initial project data.
- Citizens verify identity by mobile OTP to lodge and track complaints.
- A CERT-In empanelled “safe-to-host” security audit is completed before go-live.

3. Users, Roles & Team Management

The public side is open to all citizens; the internal side is restricted to authenticated PWD staff, with access governed by Role-Based Access Control (RBAC). A citizen can see only their own complaints, never another citizen's.

3.1 Internal Roles

Role	Typical Holder	Key Permissions
Super Administrator	IT / Nodal Officer	Full control of users, roles, divisions, projects, integrations, and settings.
Department Admin	HQ officer	Manage projects and status across divisions; manage staff; view all complaints and reports.
Division Officer	Executive Engineer	Manage projects and complaints for own division; post status updates; assign complaints.
Complaint Handler	Field / Sub-division officer	Act on assigned complaints; add notes; update status; upload resolution evidence.
Viewer / Auditor	Monitoring staff	Read-only access to complaints, projects, and reports.

3.2 Team Management

- Add any number of staff — no fixed limit on accounts.
- Each member has a name, official email/mobile, designation, division(s), and a role; roles and permissions are configurable.
- Activate, deactivate, or reassign members anytime; all staff actions are recorded in an audit log.

4. Public Module (Web & Mobile App)

4.1 Browse, Search & Filter

- View all projects with name, type (building/road/bridge/other), division, location, status, and % completion.
- Search by name, location, or project ID; filter by division, project type, and status — working together.

4.2 Project Status Tracking

- Each project page shows a progress bar with % complete and a newest-first list of dated update messages.
- Example (building): “Foundation completed” (5%) → “Plinth completed” (15%) → “First floor slab ready” (30%), each with date and time.

4.3 Lodge a Complaint

- Choose any project (in-progress or completed) and start a complaint; verify mobile number by OTP.
- Enter category, description, and location; upload one or more photos/videos (within set limits).
- On submission, receive a unique Ticket ID and confirmation (on-screen and SMS/app).

4.4 Track a Complaint (Own Only)

- Track by Ticket ID + registered mobile (OTP); a citizen sees only their own complaints.
- View the full trail — received, acknowledged, assigned, in progress, resolved — each step dated.
- See official responses and resolution evidence; get notified on every status change; give feedback once resolved.

5. Internal Module (Web & Mobile App)

5.1 Login & Dashboard

- Secure login (with OTP second factor where feasible); access by role and assigned division(s); session timeout and lockout on repeated failures.
- Role-based dashboard: new complaints, items assigned to me, overdue complaints, projects needing updates, and key counts/trends.

5.2 Complaint Management

- View all complaints in a sortable, filterable list; open to see citizen details and all uploaded media.
- Assign/re-assign to a team member; update status through the lifecycle (Section 6); add internal notes and public remarks.
- Upload resolution evidence; resolve/close; reopen if the citizen is dissatisfied. Every action is logged.

5.3 Project Management (Auto-Created from Work Orders)

- When a work order is released, the project is created automatically with its core details — no manual data entry.
- Officers enrich the auto-created project: confirm type, add milestones, contractor, expected completion, and description.
- Mark a project In Progress or Completed; the public directory reflects changes immediately.

5.4 Status Updates (Admin)

- Post an update anytime by setting the % completion and writing a short progress message; the System stamps date and time.
- Optionally attach a milestone label (e.g. “First floor ready”) and a photo.
- Full update history is kept and shown to the public, newest first; corrections by authorised users are logged.

5.5 Team & Administration

- Add unlimited staff; assign roles and divisions; edit/deactivate; reset credentials.
- Manage master data (divisions, project types, complaint categories, milestone templates) and notification/language settings.
- Configure upload limits and retention; access reports and export data (CSV/PDF).

6. Complaint Lifecycle

Every complaint moves through defined states. Citizens see a simple, transparent view; staff see the full workflow with assignment and notes. Each change is timestamped and attributed, and citizens are notified by SMS/app.

#	Status	Meaning
1	Submitted	Complaint lodged; Ticket ID issued.
2	Acknowledged	Formally received and logged.
3	Under Review	Handler is examining the complaint and evidence.
4	Assigned	Allotted to a specific field officer.
5	In Progress	Corrective action underway.
6	Resolved	Action completed; resolution recorded with evidence.
7	Closed	Closed after resolution (and feedback, if any).
8	Reopened	Re-activated if the citizen was dissatisfied.

Complaints can carry a priority and a target resolution time; overdue ones are highlighted to staff. A resolved complaint can be reopened within a defined window. All transitions are kept for audit and reporting.

7. Project Status & Progress Tracking

Progress is shown to the public as an overall % completion plus a timeline of dated update messages. Milestones structure this progress and vary by project type.

7.1 Example Milestones by Type

Type	Illustrative Milestones (in sequence)
Building	Site prep → Foundation → Plinth → Floor-wise slabs → Roofing → Finishing → Handover
Road	Survey & earthwork → Sub-base (GSB) → Base course (WMM) → Bituminous surfacing → Markings → Completion
Bridge	Foundation/piling → Substructure → Superstructure (girders & deck) → Approaches → Finishing
Other	Department-defined milestones for the work

7.2 Worked Example — Public View of a Building

Date & Time	Update Message	Completion %
02 Apr 2026, 11:20	Foundation completed	5%
28 Apr 2026, 16:05	Plinth level completed	15%
20 May 2026, 10:40	First floor slab ready	30%
12 Jun 2026, 17:15	Second floor work in progress	45%

8. Integrations & Automation

8.1 Automatic Project Creation from Work Orders

A central requirement: the moment a work order is released in PWD's work-order / works-management system, CMPTS automatically creates the corresponding project — capturing details such as work name, type, division, location, sanctioned value (if available), and reference number — and lists it in the public directory. No officer has to key it in.

- Integration via API or a scheduled feed from the work-order system (mechanism to be confirmed with PWD).
- Auto-created projects start as In Progress and can be enriched by the division officer (milestones, contractor, dates).
- If a work order is amended or cancelled, the linked project is updated accordingly.

8.2 Other Integrations

Interface	Purpose
Work-Order / Works-Management System	Automatic project creation and updates when work orders are released.
SMS Gateway	OTP verification and status-change notifications to citizens.
Mobile Push (apps)	Notifications to public and internal app users.
Email (SMTP)	Notifications to internal staff; optional citizen emails.
Map / Geolocation (optional)	Capture/display project and complaint location.

9. Non-Functional Requirements

9.1 Mobile Apps

- Native or cross-platform apps for Android and iOS, for both public and internal users, sharing the same backend as the web portal.
- Support photo/video capture and upload, push notifications, and offline-tolerant complaint drafting where feasible.

9.2 Bilingual (Hindi / English)

- Entire interface — web and apps — available in Hindi and English, switchable anytime; Devanagari must render correctly across devices.

9.3 Security & Data Protection

- HTTPS/TLS everywhere; hashed passwords; server-side RBAC; protection against OWASP Top 10 risks.
- Uploaded media validated for type/size and stored safely; comprehensive, tamper-evident audit logs.
- Aligned with DPDP Act principles for citizen data; CERT-In “safe-to-host” audit before go-live.

9.4 Accessibility, Performance & Availability

- Conform to GIGW 3.0 and WCAG 2.1 Level AA (IS 17802); responsive across desktop, tablet, and mobile.
- Fast page loads and responsiveness at peak; scales with growing projects, complaints, media, and users.
- High availability, automated backups with tested restore, and a documented disaster-recovery plan.

9.5 Usability

- Simple citizen journey — find a project, lodge a complaint, track it — in a few clear steps, with minimal jargon and government-appropriate branding.

10. Deployment, Roadmap & Acceptance

10.1 Deployment — On-Premises

- Application, database, and media storage hosted entirely on PWD-controlled servers (PWD/State Data Centre); no citizen data on a public cloud.
- Behind the departmental security perimeter; served on the PWD domain with a valid SSL/TLS certificate.
- Technology stack proposed by the vendor, subject to PWD approval and the requirements herein.

10.2 Indicative Roadmap

Phase	Focus	Key Deliverables
1	Foundation	Project directory, search & division filter, public status view; work-order auto-creation; admin updates; core team management.
2	Complaints	Citizen complaints with media, OTP, ticketing, tracking; internal workflow & notifications; mobile apps (public & internal).
3	Hardening	Bilingual completion, accessibility (GIGW/WCAG), security audit & safe-to-host, reporting & dashboards.
4	Go-Live	Data migration, training, UAT, on-prem deployment, warranty & support.

10.3 Acceptance Criteria

- All in-scope features (Sections 4–8) demonstrated and accepted via User Acceptance Testing.
- Bilingual web and mobile apps verified; work-order auto-creation demonstrated end-to-end.
- Security and accessibility evidenced, including a CERT-In safe-to-host certificate; deployed and running on PWD servers; staff trained and documentation delivered.

10.4 Open Points for Departmental Decision

1. Source work-order system and the integration method (API or feed).
2. Confirmed division list and any initial project data.
3. Max file size/formats for uploads and total storage to provision.
4. Resolution timeframes (SLAs) per complaint category, if any.
5. Whether the assigned officer's name is shown publicly; reopen window length.
6. SMS gateway / email service available within PWD; app store accounts for publishing.
7. Branding assets (State emblem, PWD logo) and the official domain.

— End of English Version —